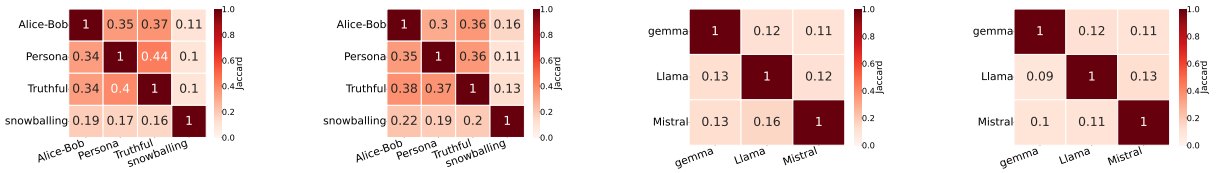


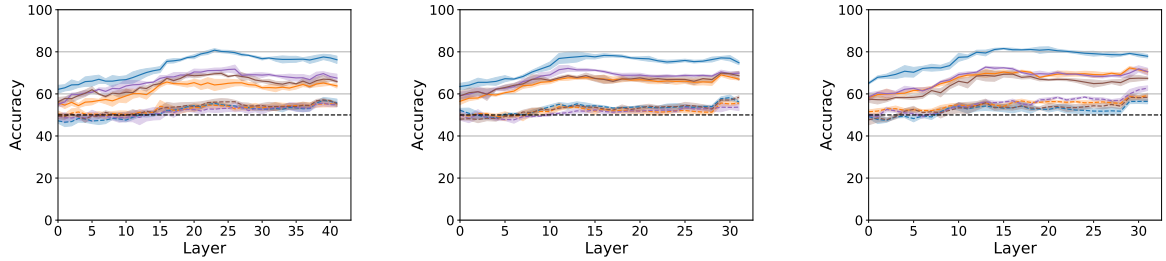
(a) Gemma HK^+ similarity between settings.

(b) Llama HK^+ similarity between settings.

(c) HK^+ similarity under Alice-Bob.

(d) HK^+ similarity under Truthful setting.

Figure 9: High-Knowledge and HK^+ differences on TriviaQA (above the diagonal) and Natural Questions (below the diagonal) between the models and settings. We can see that different models have different knowledge and different HK^+ examples in shared knowledge.



(a) Gemma.

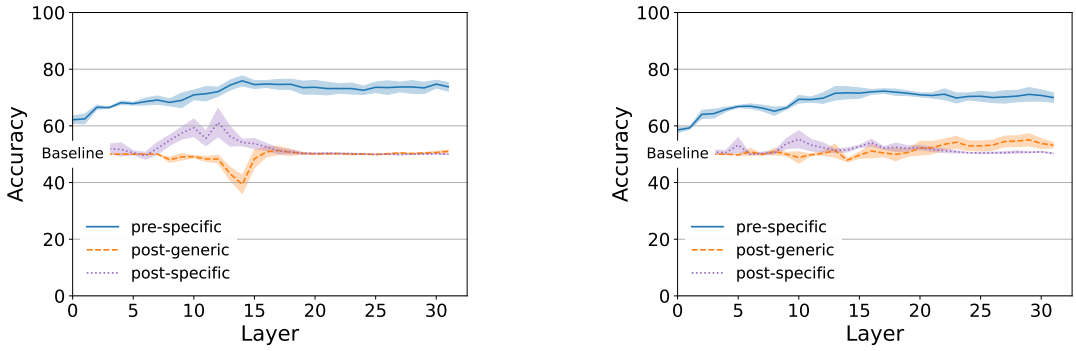
(b) Llama.

(c) Mistral.

Figure 10: Distinguishing factually correct from HK^+ using classifiers trained on generic vs. model-specific datasets on Natural Questions. We can see that specific dataset accuracy is significantly higher than generic dataset.

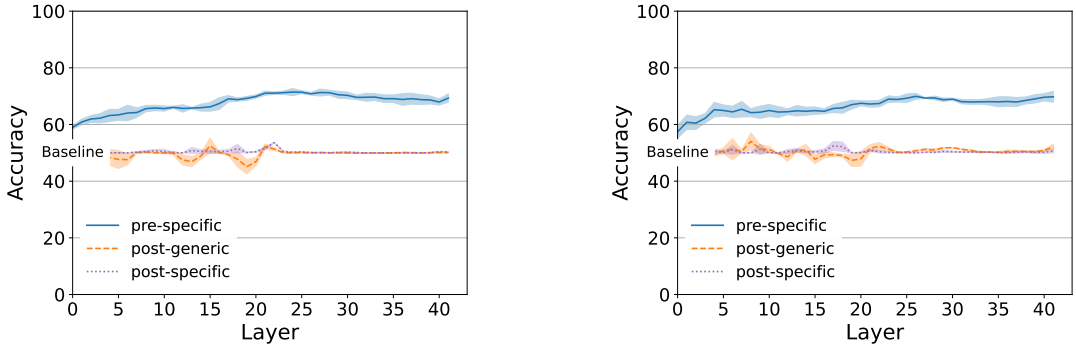
ent Snowballing. Our main objective was to show that the classifier trained under this new configuration performs similarly to the original 3-Snowballing-shot setting. Demonstrating this would suggest that variations in Snowballing configurations do not impact classifier performance.

In Figure 21 we present the performance of a classifier trained in the 5-Snowballing-shot configuration to distinguish between factually correct answers, HK^-, and HK^+. These results are comparable to those reported in the main paper in Section 4.2 for the Snowballing configurations. This consistency indicates that the 5-Snowballing-shots configuration does not affect the classifier's ability to differentiate between the various types of hallucinations.



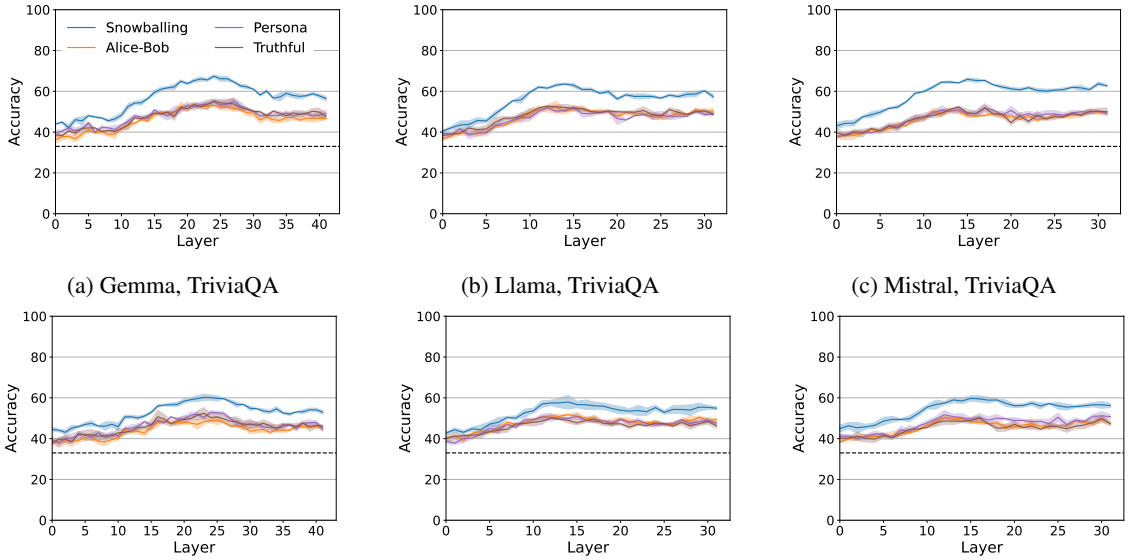
(a) TriviaQA (b) Natural Questions

Figure 11: Comparing HK^+ detection before generation using classifiers trained on model-specific pre-hallucination, generic post-hallucination, and model-specific post-hallucination examples on Llama. We can see that preemptive hallucination is effective and that generic dataset can not detect it.



(a) TriviaQA (b) Natural Questions

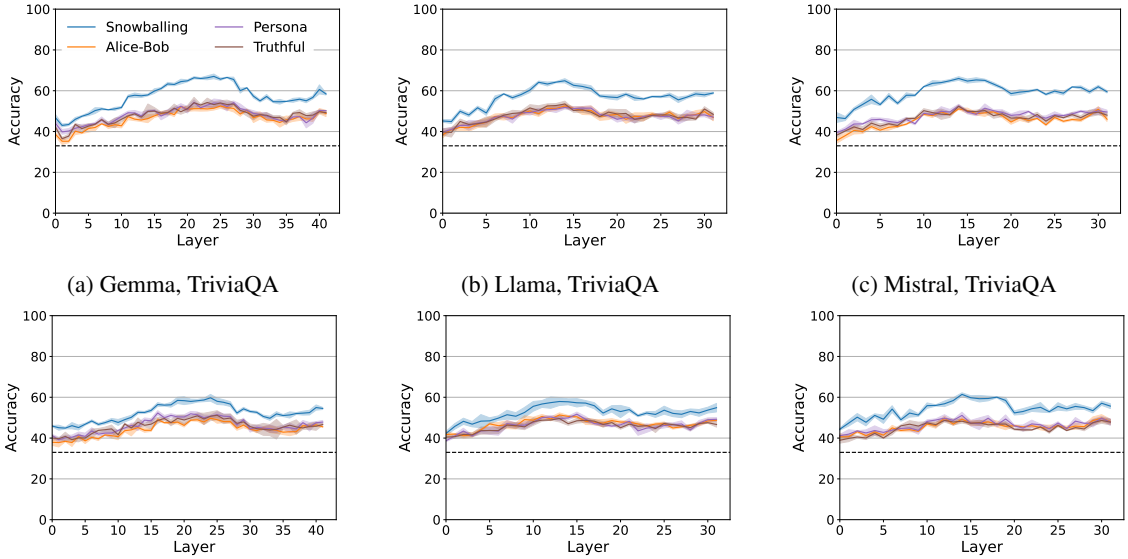
Figure 12: Comparing HK^+ detection before generation using classifiers trained on model-specific pre-hallucination, generic post-hallucination, and model-specific post-hallucination examples on Gemma. We can see that preemptive hallucination is effective and that generic dataset can not detect it.



(a) Gemma, TriviaQA (b) Llama, TriviaQA (c) Mistral, TriviaQA

(d) Gemma, Natural Questions (e) Llama, Natural Questions (f) Mistral, Natural Questions

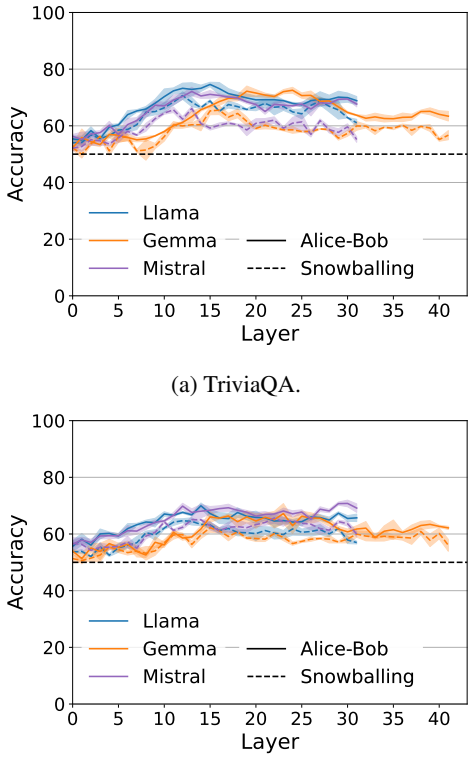
Figure 13: 3-way classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples on the MLP component. We can see that all the detection lines are well above the baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.



(a) Gemma, TriviaQA (b) Llama, TriviaQA (c) Mistral, TriviaQA

(d) Gemma, Natural Questions (e) Llama, Natural Questions (f) Mistral, Natural Questions

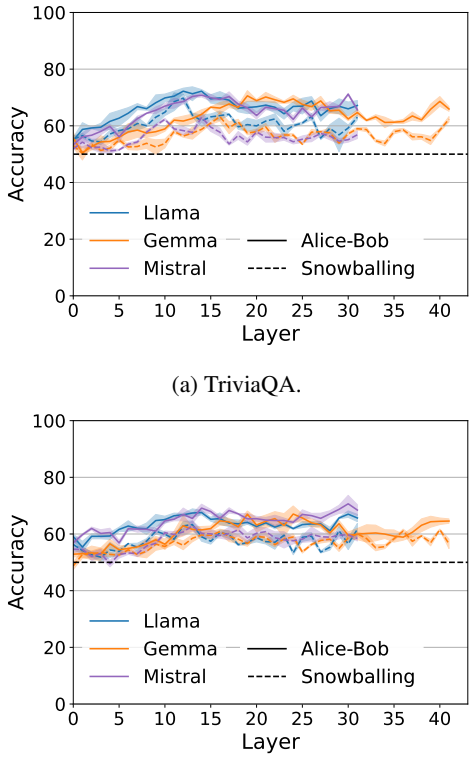
Figure 14: 3-way classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples on the Attention component. We can see that all the detection lines are well above the baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.



(a) TriviaQA.

(b) Natural Questions.

Figure 15: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or an Alice-Bob setting, and testing on the Alice-Bob setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline on MLP component.



(a) TriviaQA.

(b) Natural Questions.

Figure 16: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or an Alice-Bob setting, and testing on the Alice-Bob setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline on Attention component.